

Setting Up MongoDB

Local Installation (Optional for Beginners)

To install MongoDB on your system:

On Windows

1. Go to [MongoDB Community Download Center](#).
2. Download the MSI installer for your version.
3. Follow the installation steps and **enable MongoDB as a service**.
4. Use the terminal to run:

```
mongod
```

This starts the MongoDB server.

On macOS (Using Homebrew)

```
brew tap mongodb/brew  
brew install mongodb-community  
brew services start mongodb-community
```

On Linux (Ubuntu Example)

```
sudo apt update  
sudo apt install -y mongodb  
sudo systemctl start mongodb
```

To check if MongoDB is running:

```
mongo
```

This will open the MongoDB shell if it's installed correctly.

Using MongoDB Atlas (Cloud Setup)

MongoDB Atlas is the easiest way to get started without installing anything.

Steps to Create a Cluster:

1. Visit <https://www.mongodb.com/cloud/atlas>
2. Sign up and create a **free cluster** (Shared Tier).
3. Choose cloud provider and region.
4. Create a **username and password** for database access.
5. Whitelist your IP address or allow all IPs (`0.0.0.0/0`) for testing.
6. Get your connection string, which looks like:

```
mongodb+srv://<username>:<password>@cluster0.mongodb.net/myDatabase?
retryWrites=true&w=majority
```

MongoDB Compass (Optional GUI)

MongoDB Compass is a GUI client to visually interact with your databases.

- Download from: <https://www.mongodb.com/products/compass>
- Connect using the same Atlas URI or local URI like:

```
mongodb://localhost:27017
```

You can:

- View collections and documents
- Run queries
- Inspect schema visually

Connecting to MongoDB with Node.js

Install the official MongoDB driver:

```
npm install mongodb
```

Sample connection code:

```
const { MongoClient } = require('mongodb');

const uri = 'your_connection_string';
const client = new MongoClient(uri);

async function run() {
  await client.connect();
  const db = client.db('test');
  const collection = db.collection('students');
  const result = await collection.find().toArray();
  console.log(result);
  await client.close();
}

run();
```